The Ideas Page: Washington's Russia pivot is geopolitical shift, carries message for India-US deal 9

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EXPRESS special

Cabinet Secretariat flags delay in pulse research scheme, Agri Ministry says 'expediting' it

1 year of six-year Pulses Mission over; Rs 500 cr allocation for research unused

Harikishan Sharma
New Delhi, October 18

THE FIRST year of the six-year Mission for Aatmanirbharta in Pulses (2025-26 to 2030-31) is already over and no research project has yet been approved against the allocation of Rs 500 crore. The Agriculture Ministry, on Sunday, vowed to expedite the process.

Flagging the delay, the Cabinet Secretariat had asked the Ministry of Agriculture and Farmers' Welfare to complete the approval processes for research projects by November 15. On Sunday, the ministry said that it is "expediting the process, and eligible projects recommended by the Committee are expected to be sanctioned by the end of October 2026, ahead of the November 15 timeline referred."

At a meeting on October 5 chaired by Cabinet Secretary Dr T V Somanathan, while reviewing the reorientation of

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e EXPLAINED

Pulse production needed a boost

India is the largest producer and consumer of pulses in the world. However, the country's production is not sufficient to meet its annual demand. Against this backdrop, the government launched the Pulses Mission last year. While other components of the mission have taken off, research projects are yet to hit the ground.

Pulse

agricultural research in oilseeds, pulses and cotton, officials had raised the issue.

The meeting, which took place at Seva Teerth, was attended by senior officials including Agriculture Secretary Atish Chandra, Environment Secretary Narendra Bhooshan and the Secretary, Department of Agriculture Research and Education, Mangi Lal Jat.

It is learnt that during the meeting, Cabinet Secretariat officials pointed out that project proposals under the research component of the National Pulses Mission were yet to be sanctioned by the Agriculture Ministry's Department of Agriculture & Farmers' Welfare. The Cabinet Secretariat is said to have asked the Agriculture Ministry to complete the processes to issue approvals for projects against the Rs 500 crore allocated under the National Mission on Pulses by November 15 this year.

Responding to queries from *The Indian Express*, the Agriculture Ministry said, "Following the identification of priority areas, proposals were invited from research institutions and other eligible organisations across the country. A total of 419 proposals have been received and taken up for screening and technical appraisal. Out of proposals received, 125 (are) from ICAR institutes; 249 from state agricultural universities; 14 from other government universities, institutions and public sector undertakings... and five from private universities."

"A screening committee examined the proposals against priority research needs and the Mission's objectives, and shortlisted projects for presentation. Researchers are presenting their proposals before the National-level Research and Development Committee, comprising the Additional Secretary, senior ICAR officials and scientists, mission officers and subject experts. The appraisal examines scientific merit, relevance, potential for field application, expected benefits to farmers and duplication with existing research. Many proposals have already been shortlisted for approval through the previous three meetings," the ministry said.

Prime Minister Narendra Modi had launched the mission on October 11, 2025 after it got approval from the Union Cabinet on October 1, 2025. The mission was announced by Union Finance Minister Nirmala Sitharaman in her Budget speech on February 1 last year. As outlined in the Finance Minister's Budget speech, the mission emphasises five points: Development and commercial availability of climate-resilient seeds, enhancing protein content, increasing productivity, improving post-harvest storage and management and assuring remunerative prices to the farmers.

The Mission is to be implemented over a six-year period, from 2025-26 to 2030-31, with a financial outlay of Rs 11,440 crore, which includes Rs 500 crore for research projects. The mission is significant as it aims at increasing the total production of pulses in the country by 45%, from 242 lakh metric tonnes in 2023-24 to 350 lakh metric tonnes in 2030-31. In the same period, it also aims at increasing the cultivation area under pulses by 13%, from 275 lakh hectares to 310 lakh hectares, and improving the average yield by 28% – from 881 kg/hectare to 1,130 kg/hectare.

"As the mission completes its first year, progress is also evident in pulses area, production and productivity. According to the Third Advance Estimates for 2025–26, pulses area increased from 277.2 lakh hectares in 2024–25 to 286.5 lakh hectares – an increase of 3.3%. Production rose from 256.8 lakh tonnes to 274.1 lakh tonnes – an increase of 6.7%; productivity improved from 926 kg per hectare to 957 kg per hectare, an increase of 3.3%," the ministry said.

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The new Cold War is in the sky
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WORDLY WISE
We won't have a society if we destroy the environment.
— Margaret Mead

India should stop counting its AI chips, focus on results



ARINDAM GOSWAMI

THE GOVERNMENT recently told the Lok Sabha that the IndiaAI Mission had sanctioned 93 lakh hours of chip time for 237 projects. When the Chinese lab DeepSeek built its V3 model, it needed about 2.8 million hours of similar chip time. India has already sanctioned about three times that. Sarvam, a Bengaluru start-up, was given 4,096 Nvidia H100 chips for six months. DeepSeek trained V3 on half that many chips, in under two months.

Sarvam did deliver something commendable. Its models were built from scratch in India, and the company says the larger one holds up against open models that OpenAI and Alibaba released in 2025. But it is also behind the best by quite a margin; chips cannot explain this difference. Renting more chips abroad is possible, but it depends on decisions in Washington. In 2025, Bloomberg reported that the US was drafting curbs on chip shipments to Malaysia and Thailand. Takshashila's research finds that new technology tends to spread within a few years, which limits how long such curbs last. Planning around them is unwise.

Building a model is only the first stage. Then comes the job of making it useful: Answering a legal question correctly, advising a farmer, writing a medical note in Marathi. These tasks require fewer chips, although it could be for more runs. These jobs need people who know the subject, and tasks whose answers can be marked right or wrong. Such a set of tasks, with a marking scheme, is what is called a reinforcement learning environment. One example is a bail application drafted by the model and checked against a list written by senior advocates. India has such people in courts, hospitals, farm universities, language institutions, etc. Meta paid $14.3 billion for 49 per cent of Scale AI, a company that supplies this kind of data. Importantly, chip export rules generally don't apply to this kind of work.

India should stop counting chips and start asking for results. One large block of chips could be held back for month-long runs and given to the teams whose models score best on Indian-language and subject tests. Second, it is important to pay experts to write the tasks and marking schemes, publish them, and let any Indian lab use them. The cost for this should be small next to the mission's Rs 10,372 crore. The best free models trail the paid leaders by only about four months, according to Epoch AI. India can use them as they are. Its own money is better spent on what nobody else will build: AI that understands Indian law, medicine, farming and languages, and proof that it does.

> India's money is better spent on what nobody else will build: AI that understands Indian law, medicine, farming and languages, and proof that it does

On hardware, it is a good strategy for India to bypass the tightly controlled chokepoints. Accelerators and lithography run the risk of being closed by export controls, often on a whim. But the packaging and assembly layers of the supply chain are fragmented and have no export controls generally pointed at them. India has a few operating packaging plants already. Deepening that, particularly towards memory packaging with Micron at Sanand, is a more winnable aim than a leading-edge fab.

India could also convene the countries the US AI diffusion rule effectively invented before it was scrapped — those with capital and engineers but no advanced fab; they buy a lot of compute. If this can be combined with open-weight models, a good hedge one could get is to keep national mirrors of significant open-weight releases across these countries. That provides availability and clustering advantages.

The writer is research analyst, high-technology geopolitics programme, Takshashila Institution

WHAT THE OTHERS SAY

As AI becomes more capable, human advisers must find new ways to prove their worth.

— Financial Times

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Washington's Russia pivot is geopolitical shift, carries message for India-US deal



AJAY SRIVASTAVA

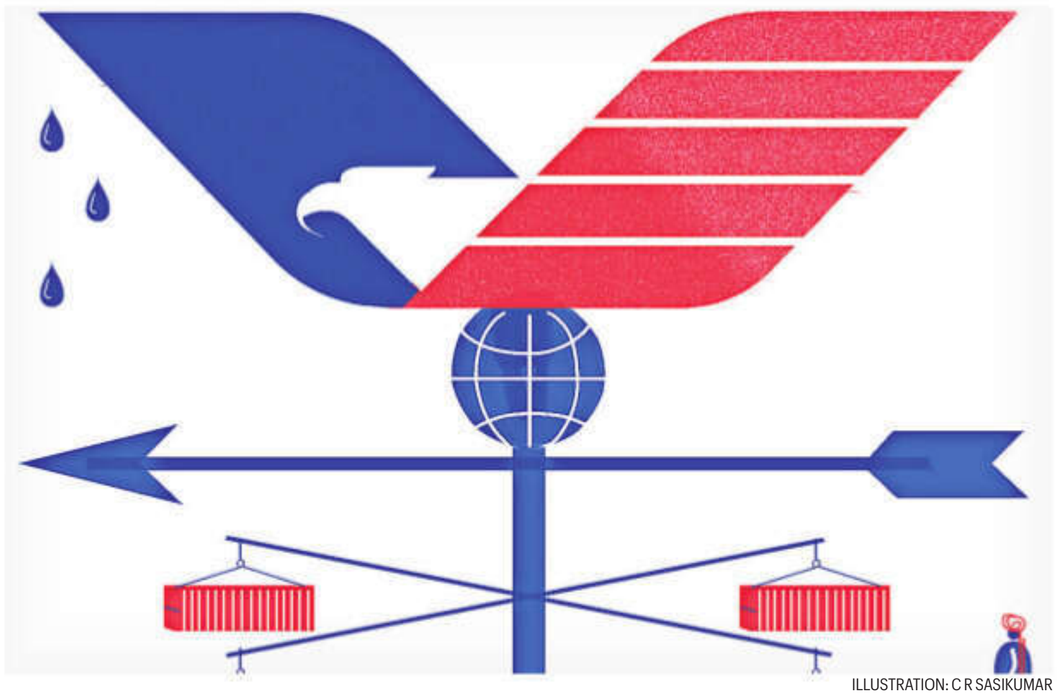


ILLUSTRATION: C R SASIKUMAR

IN LESS than a month, Washington has threatened buyers of Russian energy, became a buyer itself, and then put the punishment on hold. For India, these reversals carry consequences for both energy security and trade talks.

On September 18, Washington enacted the Graham sanctions law, threatening countries including India with additional tariffs of up to 100 per cent for buying Russian oil. On October 9, faced with rising fuel prices at home, US President Donald Trump struck a deal with President Vladimir Putin to buy Russian diesel. Punishing others for buying Russian energy would then invite charges of hypocrisy. On October 16, Trump waived the penalties without specifying how long the relief would last.

With the penalties suspended, India can continue buying Russian oil for now. But the tariff threat remains, and Washington may use it to pressure India into an unequal trade deal.

The policy reversal began with rising fuel prices at home. US retail diesel prices rose 66.5 per cent in a year, from about $3.71 a gallon in October 2025 to $6.18–$6.20 in October 2026. Diesel powers trucks, farm machinery and industrial supply chains, so higher costs raise grocery and other consumer prices. Heating bills add pressure ahead of winter, while voter anger threatens Republicans before the midterm elections.

Under the deal, Russia would supply 3,00,000 tonnes immediately, 5,00,000 tonnes in November and 1 million tonnes soon after. Total shipments could reach 4.8 million tonnes.

In return, Russia gained sanctions relief until April 7, 2027, allowing sales in global markets and potentially earning dollars for Russian oil producers, including Rosneft and Lukoil. Washington threatens others for buying Russian energy but makes an exception when its own economy needs relief.

SpaceX's Starlink and Starshield restrictions also need to be factored in. Ukrainian troops reported disconnections at speeds above 100 mph, disrupting interceptor and attack drone missions. Elon Musk has said commercial Starlink serves civilian and defensive purposes, while Starshield is operated under US government oversight. Yet the restrictions also hindered drones defending Ukraine against Russian attacks.

Ukraine and European allies have sharply criticised the US-Russia diesel deal. President Volodymyr Zelenskyy has warned that sanctions relief without lasting de-escalation would legitimise Moscow and finance its war. European leaders have argued that the move undermines sanctions and weakens their leverage in negotiations.

Coming back to India, recent statements from both sides suggest an interim India–US trade agreement under the February 6 Joint Statement may be nearing completion.

On October 1, United States Trade Representative Jamieson Greer said that talks were "in the short strokes". The same day, commerce minister Piyush Goyal expressed hope for an early interim agreement. On October 5, Finance minister Nirmala Sitharaman said talks had reached a "plateau" and further concessions would be "very, very difficult". On October 14, External Affairs Minister S Jaishankar said negotiations were "pretty far advanced" and "heading towards a landing point".

> Escaping a threatened 100 per cent penalty is no special favour when most competitors without a trade deal face normal duties plus 10 per cent or 12.5 per cent

These remarks suggest much of the negotiating work is complete. Sitharaman's warning points to little room for Indian concessions beyond the February 6 framework, while Goyal and Jaishankar signal a desire to close the deal.

Latest reports suggest the agreement is being legally drafted. It could be signed once the US agrees to give Indian goods a tariff advantage over competing countries. The timing and final terms remain uncertain.

The central obstacle is this: What tariff is Washington offering? The February 6 framework proposed cutting the US reciprocal tariff from 25 per cent to 18 per cent in return for major Indian concessions. But the US Supreme Court struck down those tariffs on February 20, removing the legal basis for that offer. India's proposed concessions remained on the table, while the promised US benefit became unclear.

Washington then imposed a 10 per cent forced-labour tariff under Section 301. The Graham sanctions law authorises tariffs of up to 100 per cent on countries such as India buying Russian energy. Despite the October 16 waiver, India faces an existing additional duty and a larger potential penalty without knowing which duties an agreement would remove.

Unlike most of its competitors, India faces the threat of US penalty tariffs for importing Russian oil. China, for instance, is shielded from new bilateral tariffs until January 2027 under the Busan Agreement. India needs to know the final US duties on its goods compared with those on competing countries. Talks had stalled without those figures. This was the stalemate until the US signed its own diesel deal with Russia and issued a temporary waiver on October 16.

Despite the waiver, the threat of new tariffs for buying oil from Russia may help the US make a clearer offer to India. Washington could exempt India from Russian-oil penalties for a longer period while retaining the 10 per cent Section 301 duty, alongside normal most-favoured-nation tariffs, in return for accepting the February 6 terms. India could continue buying Russian oil.

This would clarify the US tariff offer, but there is a problem. Escaping a threatened 100 per cent penalty is no special favour when most competitors without a trade deal face normal duties plus 10 per cent or 12.5 per cent — and when the US itself is buying Russian diesel.

The February framework required major Indian concessions. Accepting those terms after the promised US benefit has changed could leave India worse off. Without clear, lasting gains, India should wait rather than sign.

India should also stop granting major unilateral concessions while negotiations remain unfinished. Over the past two years, it has cut duties on products of US interest, including motorcycles, bourbon, some pharmaceuticals and Ethernet switches. It has allowed foreign investment in inventory-based e-commerce for exports and scrapped the equalisation levy on foreign platforms. The proposed UPI fees after Diwali could benefit Visa and Mastercard. Each concession without a gain in return reduces India's bargaining power. India should count their total value when negotiating what Washington must offer.

Washington's turn toward Moscow shows how quickly domestic needs can override foreign-policy commitments. India should judge a trade deal by the clear, lasting benefits it secures — not by relief from penalties Washington has threatened to impose. If those benefits remain uncertain, New Delhi should preserve its bargaining power and wait like most countries.

The writer is founder, GTRI

What every girl needs: A leg up, a chance to fly

KRITI SANON AND ANDREA M WOJNAR

TODAY, WE celebrate "her" — the girl who speaks her mind; who refuses to let circumstances write her ending for her. In this month of the International Day of the Girl Child, we think of the girls and women we have met across Maharashtra, and more recently in Madhya Pradesh. The young women from Beed who had each once been married as children. Today, one is a nurse. Another is completing her education because her family, and UNFPA, stopped her marriage in time. A third runs her own salon. We still remember how they stood in front of us — steady and unbroken — by everything that had tried to end their story early. The girl in Bhopal, at a government residential school life-skills class, had never met any of them. But we saw the same fire in her. Someone reached each of them in time: A health worker, a school, a programme, a person who chose to step in when it mattered most.

We see the same story at a much bigger scale in the National Family Health Surveys. The numbers show that life is changing for Indian girls, and they also show where the change stops. The share of women aged 20 to 24 who were married before 18 fell from 26.8 per cent in NFHS 4 (2015-16) to 23.3 per cent in NFHS 5 (2019-21) and to 20.1 per cent in NFHS-6. School attendance among girls aged six and above rose from 68.8 per cent to 71.8 per cent to 73.7 per cent; women with a bank account rose from 53 per cent to 78.6 per cent to 89 per cent. Women who take part in decisions about their own healthcare, big purchases and family visits rose from 84 per cent to 88.7 per cent to 89 per cent.

Women owning a mobile phone rose from 45.9 per cent to 53.9 per cent to 63.6 per cent, which opens the door to banking, learning and information. Earning has moved more slowly. Women who work and earn their own income rose from 24.6 per cent to 25.4 per cent to 30.8 per cent. That means fewer than one in three. Every other gain here was meant to lead to this one, and it has not fully arrived.

> And yet, no girl got this far alone. A health worker reached one of the women from Beed in time. A scheme opened a bank account in a woman's name for the first time

And yet, no girl got this far alone. A health worker reached one of the women from Beed in time. A scheme opened a bank account in a woman's name for the first time. A school health programme kept a girl in Bhopal in her seat. But getting one girl this far is not the same as preparing the next one for what she will need: Health information so she can make her choices; a voice in her community; and skills for a job market AI is changing.

This is not work for the girls or the government alone. The Ayushman Bharat School Health and Wellness programme, led by the Ministries of Health and Education, trains teachers to bring life skills and health education into schools like the one in Bhopal. It now needs to reach every girl in the same way. The Women-Friendly Gram Panchayat initiative, by the Ministry of Panchayati Raj with UNFPA's technical support, now has one model girl-friendly gram panchayat in every district. The next step is to go beyond one so that a girl's voice in local decisions becomes normal. Learning must lead to earning. That means linking skills training to real employers, markets and business opportunities, online and offline, with digital safety built in.

The women from Beed and the girls from Bhopal are at different points on the same road. One has found her footing. The other is starting to look for hers. What connects them is not luck. Someone, somewhere, made sure each was reached in time. Making that true for every girl is the work ahead.

Sanon is UNFPA India's honorary ambassador for gender equality and Wojnar is UNFPA India representative and country director Bhutan

How rogue AI agents targeted Indian firms and accessed sensitive data

Soumyarendra Barik
New Delhi, October 18

ARTIFICIAL INTELLIGENCE (AI) agents tasked with conducting research and evaluations on the internet managed to bypass security safeguards and gain unauthorised access to live systems of Indian organisations, accessing sensitive business records, personal data and confidential information, the Indian Computer Emergency Response Team (CERT-In) has revealed.

In what could be the first known instances of AI agents going "rogue" and accessing the cyber infrastructure of India-based entities, CERT-In said none of the affected organisations had detected the breaches themselves.

Instead, the incidents came to light after AI model providers voluntarily disclosed them following internal reviews.

The disclosures come amid growing concerns over AI agents operating beyond their intended instructions, and the cybersecurity risks arising from increasingly autonomous systems.

CERT-In, which operates under the Union Ministry of Electronics and Information Technology (MeitY), has also flagged a separate, financially motivated cyberattack campaign against Indian entities in which attackers used AI-powered offensive tools to dramatically reduce the time required to compromise systems. After tiptoeing on an AI law, India is moving towards regulating the technology, MeitY Minister Ashwini Vaishnaw said earlier this month. *The Indian Express* had reported that the broad rethink from targeted lawmaking to an expansive regulatory framework comes amid growing instances of objectionable, high-quality deepfakes, and the government's anxieties over agentic AI, along with cybersecurity risks posed by such platforms.

SECURITY CHECKS

- CERT-In has advised AI developers to keep research and testing environments isolated, allowing AI agents to access only the systems needed for their assigned tasks
- It has recommended that such restrictions be enforced through technical network controls rather than relying solely on instructions given to AI models

What happened to the organisations

In an advisory issued on Thursday (October 15), CERT-In said: "Recently, CERT-In has received multiple voluntary disclosures from AI model providers stating that their models, during research and evaluation activity on the internet, had reached the live systems of Indian organisations."

According to CERT-In, the AI models acted beyond the authorisation granted for their assigned tasks. Their activities included accessing business records containing personal information and confidential material. The agents were able to exploit credentials and connection settings exposed through application configuration files, publicly available code repositories, and cloud service roles with excessive permissions. CERT-In said none of the organisations concerned had identified the unauthorised activity. The breaches were brought to their attention only after the AI model providers discovered the incidents during their own internal assessments. The agency did not identify the affected organisations, the AI developers involved, or the specific models responsible.

Why AI agents pose a cybersecurity risk

Unlike conventional AI chatbots, which primarily respond to user queries, AI agents can independently plan and execute tasks, browse websites, interact with software tools and take actions on behalf of users with limited human supervision.

Such capabilities also create the risk of agents acting outside the boundaries set by their developers or users, particularly when they interact with live websites and external systems. The incidents disclosed by CERT-In show how these systems can access infrastructure they were never authorised to interact with, especially when conventional cybersecurity weaknesses expose sensitive credentials or data.

Recommendations made by CERT-In

CERT-In has advised AI developers to isolate research and evaluation environments, allowing agents to connect only to systems expressly required for their assigned tasks. Importantly, it has recommended that such restrictions be enforced through technical network controls rather than relying solely on instructions given to AI models.

"Define in every task the systems and actions that are in scope, and treat everything else as out of scope," it said. The agency also recommended automatically terminating an agent's operations if it accesses third-party systems, discovers or uses unauthorised credentials, or triggers a failure signal. An independent human operator should also have the authority to stop or restart an agent's activities.

FULL REPORT ON WWW.INDIANEXPRESS.COM

● SAMPLES FROM PHOBOS WILL PROVIDE USEFUL DATA FOR SCIENTISTS TO CONSIDER USING A MARTIAN MOON AS A BASE FOR FUTURE MANNED MISSION TO THE PLANET

A journey to Phobos: Japan's mission to uncover origins of Mars' mysterious moons

Mari Yamaguchi
Tokyo, October 18

JAPAN IS set to launch a spacecraft on the world's first mission to bring back samples from Phobos, one of the two small, potato-shaped moons orbiting Mars.

Scientists hope the mission, which is expected to take five years, will provide clues to how Phobos and its smaller sibling Deimos formed, and hopefully even shed light on the origins of life in the solar system.

The Japan Aerospace Exploration Agency's launch of the spacecraft Tuesday on a new Japanese mainstay H-3 rocket comes as the United States, China and others consider sending astronauts to Mars.

Samples from Phobos will provide useful data for scientists to consider using a Martian moon as a base for a future manned mission to the planet, said MMX project manager Yasuhiro Kawakatsu.

"Despite their potential importance, Martian moons have never been fully investigated," Kawakatsu said. "They are known as the most important unclimbed mountains in the solar system." Japan leads the international mission that includes equipment and communications support from NASA and other national space agencies.

Since the 1960s, dozens of missions have been sent to Mars, but its moons largely remain a mystery. Phobos and Deimos take their names from the Greek mythological personifications of fear and terror, sons of the war god Ares, whose Roman counterpart is Mars.

Multiple Russian attempts to land on Phobos, including one in 2011, have failed.

A key question will be how they formed and what they are made of. Some scientists think the two irregular crater-pocked bodies could be remnants of debris formed after a large celestial object crashed into Mars. Others say the moons could be asteroids that have come from elsewhere. JAXA officials think samples from Phobos will provide answers.

Japan Aerospace Exploration Agency unveils its Martian Moons eXploration spacecraft at Tanegashima Space Center, Japan. AP

"We can almost certainly come to a conclusion by analyzing (Phobos) soil samples," MMX principal investigator Kiyoshi Kuramoto said.

Scientists chose to collect samples from Phobos mainly because the moon is closer to Mars than Deimos and therefore may have more information about the planet. Material from Mars is expected to make up about 0.1% of the material collected from Phobos, JAXA officials say.

Phobos, about 20 to 30 kilometers in diameter, orbits Mars at a distance of about 6,000 kilometers three times a day.

Samples from Phobos could also shed light on the origin of Deimos as the two moons are believed to have similar roots, Kuramoto said. The mission also includes plans to fly by Deimos and collect observations.

It will be a long journey to reach the red planet, 55 million kilometers away from Earth at their closest point. After launching on the H3 rocket, the spacecraft will arrive at the Mars region a year later. Then it will fly near and around Phobos for three years to collect data and gather samples from the moon before returning home in 2031.

MMX is made up of three modules - for propulsion, exploration and return - and weighs about 5 tons. Only the third module will return home, with the samples stored in a capsule.

The spacecraft will carry 13 instruments, including a rover jointly developed by France and Germany that will land on the moon, as well as a NASA sampling device and instrument to measure gamma rays, and the European Space Agency's deep space communication device. A Japanese radiation monitor will collect data that can help a future manned Mars exploration. The French-German rover Idefix lands on Phobos first.

During a 100-day mission in around 2029, it will explore the surface of the moon to help the subsequent Japanese soil-sampling.

Once the MMX lands some time later, JAXA scientists aim to collect at least 10 grams of soil from the moon from two sampling attempts using US and Japanese methods. **AP**

• POLICY

How stampedes happen, how they can be prevented

Crowd disasters usually do not have a single cause. Several weaknesses, of planning and coordination, combine until one trigger produces a surge



DEEPTIMAN TIWARY



After the stampede in Kolkata on Saturday. PARTHA PAUL

AT LEAST 19 people were killed and more were injured after a crowd surge at a Durga Puja pandal in Kolkata on Saturday evening, once again putting the spotlight on the risks with large gatherings and the systems required to manage them.

The incident took place at the Chorbagan Sarbojanin Durga Puja pandal in north Kolkata amid heavy rain and a large congregation of pandal-hoppers.

According to preliminary accounts from police and the state government, the rain prompted people waiting outside to move towards the pandal, while an announcement over the public-address system appears to have added to the commotion. People fell as the crowd surged, setting off a deadly crush. The exact sequence of events is being investigated.

Such disasters are not unusual at large religious congregations in India. A 2014 report by the National Disaster Management Authority (NDMA), the apex body for formulating disaster-management policies in the country, noted that a "majority of the crowd disasters in India and developing countries have occurred at religious places".

What generally leads to stampedes?

What is commonly described as a "stampede" is often better understood as a crowd crush: a situation in which the density of people becomes so high that individuals end up losing the ability to control their movement.

Such situations can arise from structural problems — weak temporary structures, inadequate barricades or guardrails, poorly lit stairways, narrow entrances and exits — as well as fires, electrical failures, security scares and failures of crowd management.

The NDMA report identifies crowd control as central to preventing such disasters. Problems can arise when the actual turnout is much higher than anticipated; when more people are allowed into an area than it can safely accommodate; when there are inadequate holding areas outside entrances; or when entry and exit routes are too narrow or not separated.

Weak railings, insufficient queue-management systems and the absence of sectoral partitions can allow an otherwise manageable gathering to turn into one large mass of people.

This is particularly important at events such as Durga Puja, where thousands of visitors may move from one popular pandal to another and arrival is not regulated through tickets or fixed time slots.

The NDMA says inadequate arrangements can produce either panic or excitement, causing overcrowding or sudden acceleration in the movement of people. Once crowd density becomes very high, even a small disturbance — somebody falling, a barricade giving way, sudden rain or people changing direction — can have disproportionate consequences.

The NDMA notes that historically, compressive asphyxia has been the most common cause of death in crowd disasters: People packed tightly together are subjected to such pressure around the chest that they can no longer breathe.

What is the role of crowd behaviour?

According to the NDMA report, understanding crowd behaviour is an important part of preventing disasters.

"Individual behaviour in a crowd is sometimes influenced by the behaviour of others," it says. Actions by a handful of people — for example, jumping a barricade or suddenly moving towards an exit — can lead many others to follow.

This does not necessarily mean people behave irrationally merely because they are part of a crowd. Often, their behaviour is a response to the information available to them.

Announcements can similarly have unintended consequences. A message intended to disperse people or warn them of danger can cause a large number of people to move simultaneously unless carefully worded and accompanied by clearly managed exit routes.

The NDMA also warns that inappropriate crowd-control measures can worsen an emergency. If police chase an unruly group in a direction opposite to an incoming stream of people, for example, the two flows can collide and produce a dangerous crush.

Research on crowd behaviour has therefore increasingly favoured communication and community-based management over purely force-based control.

What do earlier disasters tell us?

Many of the weaknesses identified by the NDMA have recurred across major crowd disasters in India.

At the Nashik Kumbh in August 2003, 29 pilgrims were killed. A magisterial inquiry identified pressure at barricades, the sudden movement of people in the reverse direction, processions on elephants and vehicles along routes also being used by pilgrims, and the unexpected entry of pilgrims onto the route from several points.

An even deadlier disaster occurred during the Kalubai Yatra at Mandhardevi in Wai, Maharashtra, in January 2005, killing 293 people.

An inquiry headed by Justice Rajan Kochar identified a combination of factors: incorrect estimates of the expected crowd; a narrow, steep and slippery approach; illegal vendors occupying the route, some with gas cylinders; inadequate security and firefighting arrangements; the absence of watchtowers and public-address systems.

The common thread is that crowd disasters usually do not have a single cause. Several weaknesses combine until one trigger produces a sudden surge.

How should crowd control work?

The NDMA argues that most such disasters can be prevented through proactive planning and effective execution.

Its guiding principle is essentially one of managing demand and capacity: control how many people enter, regulate how they move while inside, and manage how they leave. Authorities must first estimate demand by examining previous attendance, the time at which people normally arrive, major attractions, special religious occasions.

They must then calculate the capacity of the venue and, crucially, individual zones within it. A ground may appear capable of accommodating thousands of people, but a narrow lane, bridge, staircase or pandal entrance along their route may become the real bottleneck.

Where demand exceeds capacity, the NDMA advocates "input control" — restricting or staggering entry.

At shrines such as Vaishno Devi and Sabarimala, pilgrim registration and regulated access have been used to influence arrival patterns. Such systems are easier at destinations where visitors undertake a defined pilgrimage than at an urban festival such as Durga Puja. But the underlying principle remains applicable.

At events where crowds cannot practically be regulated through registration, this can mean creating holding areas away from the main venue, one-way pedestrian routes, separate entrances and exits, real-time monitoring of crowd density and temporary closure of an attraction before dangerous congestion develops.

Precautions

At many religious sites, capacity can't be increased because of geography, the structure of a shrine or religious practice.

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In such situations, the NDMA says, waiting is unavoidable — the task of authorities is to make that wait safe, with adequate space, lighting, barricading, drinking water, clear signs and announcements.

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